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Paleomagnetism of Khatyrka and Maynitsky superterranes, Koryak highlands, far eastern Russia

Alexi Didenko [a], William Harbert [b] and Anatoly Stavsky [c]

[a] *Institute of Physics of the Earth, Academy of Sciences, Moscow, Russia*
[b] *Department of Geology, University of Pittsburgh, Pittsburgh, PA 15260, USA*
[c] *Aerogeologia, Ministry of Geology, Moscow, Russia*

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ABSTRACT

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We have completed a paleomagnetic reconnaissance study of sedimentary and volcanic extrusive rocks collected from two major tectonic zones in northeastern Russia. Paleomagnetic sites were sampled within the fault-bounded structural units of the Khatyrka and Maynitsky superterranes and an overlap sequence of the Khatyrka superterrane. These sampling localities were chosen to allow both within-site and between-site fold tests. Stepwise thermal demagnetization within the temperature range 200–640°C showed a characteristic linear demagnetization path between thermal demagnetization steps of 400°C and 530°C. For thermal steps above 550°C, the magnetic intensity of many samples began to increase rapidly with magnetic directions, which were random between heating steps, suggesting the formation of new magnetic phases in these samples. Paleomagnetic samples collected from basalts and sediments of the Khatyrka superterrane and basalts and gabbros of the Maynitsky superterrane pass fold tests and show significant poleward motion of these superterranes since the formation of their rocks. The observed paleomagnetic paleolatitudes between 24°N or S and 32°N or S can be compared with expected paleolatitudes of 57°N to 79°N. Paleomagnetic results from sites collected from overlapping Senonian rocks pass a fold test at the 99% confidence level and give a pole position not significantly different from that expected from the apparent polar wander path for the Eurasia or North America plates, suggesting that these sedimentary units overlapping the Khatyrka superterrane were deposited along the ancient northeast margin of the Eurasian plate. The declination, in stratigraphic coordinates, shows a maximum clockwise rotation of about 20° when compared with the Eurasian APWP.

Introduction

During the last ten years new geological and geophysical investigations have shown the existence of tectonostratigraphic terranes along the margins of both the North America and Eurasia plates. Most of these studies have used paleomagnetic data to demonstrate that tectonostratigraphic terranes, presently located in southern Alaska or northeastern Russia, originated thousands of kilometers south of their present location and were accreted after significant poleward transport had occurred (Coe et al., 1985; Coney and Jones, 1985; Khramov, 1988; Sokolov, 1988; Harbert, 1990; Stavsky et al., 1990; Didenko et al., 1991). Within the Koryak region, the area of northeastern Russia bordering the northwestern Bering Sea (Fig. 1), models of horizontal motion of tectonostratigraphic terranes, which comprise this region, are presently based on geological data (Sokolov, 1988; Stavsky et al., 1990). Because of the lack of paleomagnetic studies from the terranes of the Koryak region these tectonic models should only be considered tentative. Presently, there are no high-reliability paleomagnetic data from the Koryak region, even after 50 studies (Khramov, 1988). In order to accurately deter-

Correspondence to: Dr. W. Harbert, Department of Geology and Planetary Sciences, University of Pittsburgh, 321 Old Engineering Hall, Pittsburgh, PA 15260, USA.

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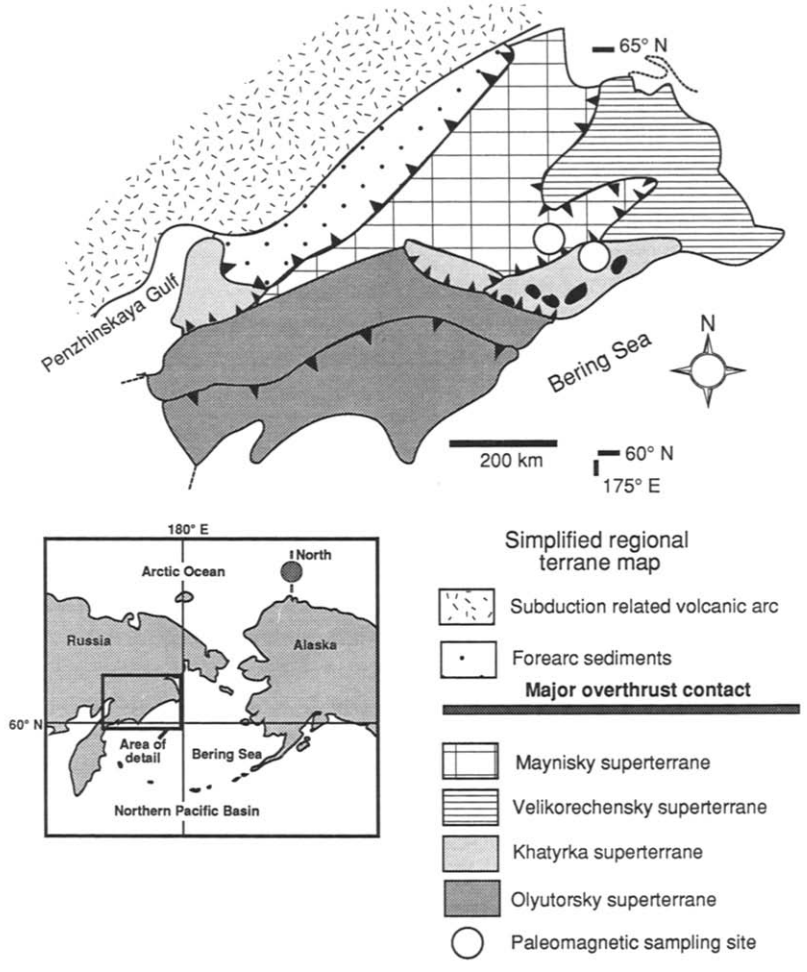


Fig. 1. First-order tectonic sketch map of the superterranes of northeastern Russia (Sokolov, 1988). The small insert map in the lower right hand corner shows the relationship of this area, the Koryak region, to the overall geography of the Bering Sea region.

mine the paleogeography of this region, using paleomagnetism, we completed a reconnaissance paleomagnetic sampling of an overlap sequence bounding tectonostratigraphic terranes of the Koryak region and reconnaissance sampling of two of the terranes themselves.

Brief geological overview

Most outboard of the terranes of northeastern Russia is the Olyutorsky superterrane, which is located along the northwestern margin of the Bering sea. To the northwest of this superterrane are three major tectonic superterranes, which have been distinguished in the geological interior of the Koryak region (Ruzhentsev et al., 1982; Sokolov et al., 1987; Sokolov, 1988), and are bounded by SW–NE-striking thrust faults. Progressing to the northwest from the Olyutorsky superterrane is the Khatyrka superterrane, bordered by the Velikorechensky superterrane to the northeast and the Maynitsky superterrane to the northwest (Figs. 1 and 2).

Each of these superterranes may also be subdivided into smaller units. Figure 2 shows a somewhat more detailed division of these larger superterranes (Ruzhentsev et al., 1982; Sokolov et al., 1987; Sokolov, 1988). Complex geology is representative of each of these smaller terranes (Fig. 3). Because we are unsure as to whether individual units making up the superterranes represent distinct terranes or complexly deformed portions

of a depositionally related unit, we shall refer to smaller zones as "structural units" of the larger superterranes.

The Khatyrka superterrane consists of a series of fault-bounded structural units, which appear to have been sutured to this region of Eurasia during the upper Mesozoic (Ruzhentsev et al., 1982; Sokolov, 1988). The Ryelyavaamski structural unit occupies the lowest stratigraphic position in the paleomagnetic sampling region and is below the Ekonay, Kokuyskyi, and Iomraut structural units of the Khatyrka superterrane. An overlapping coarse clastic sequence of Senonian to Paleocene rocks locally covers these other structural units (Fig. 3).

The Maynitsky superterrane is interpreted to represent a highly deformed, thrust-faulted, remnant of an island-arc complex. A 400-m-thick stratigraphic section of Callovian to Hauterivian volcanoclastic sediments containing fragments of boninite and tholeiite series igneous rocks is observed in the northern part of the zone. The southern part of the Maynitsky superterrane consists of Kimmeridgian to Hauterivian graywackes and olistostromes, which have been interpreted to represent an ancient subduction complex associated with an island-arc (Figs. 1 and 2).

A complete ophiolite section has been studied in the central part of the Maynitsky superterrane, within the Yagyelnyi tectonic zone. This zone consists of faulted blocks of layered gabbros, ultrabasics, sheeted dikes and basalt flows. These ophiolites have been interpreted to represent a fragment of the marginal basin crust on which the island-arc rocks of the Maynitsky superterrane are thought to have been deposited (Ruzhentsev et al., 1982; Sokolov et al., 1987).

The Velikorechensky superterrane consists of a system of thrust fault-bounded, tectonic slices of Upper Jurassic to Cretaceous sedimentary

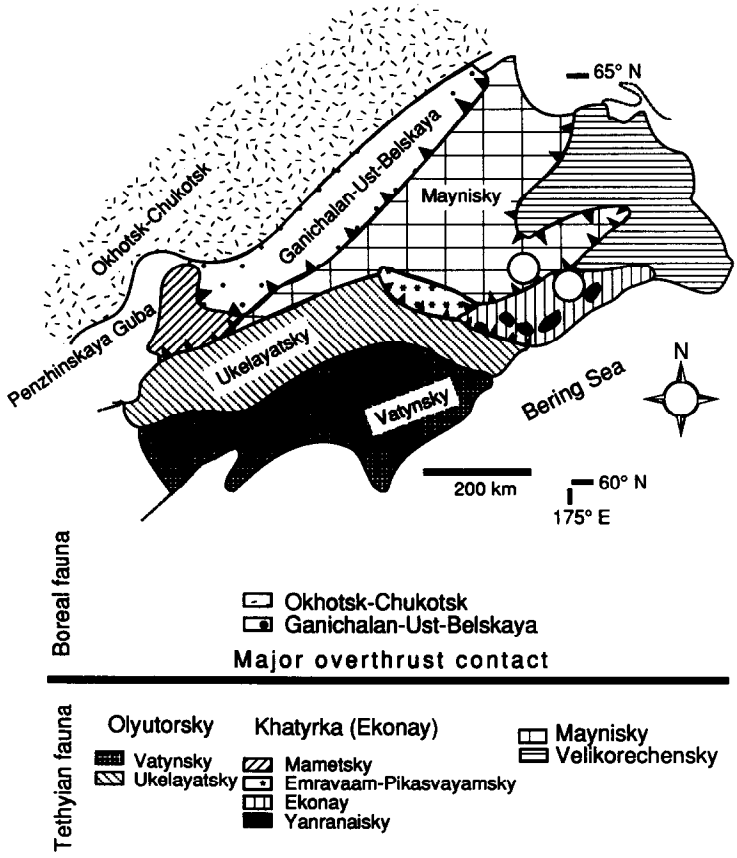


Fig. 2. Tectonic sketch map of the Koryak region and surrounding areas (Sokolov, 1988). The large open circles show the positions of the regions of our reconnaissance paleomagnetic study.

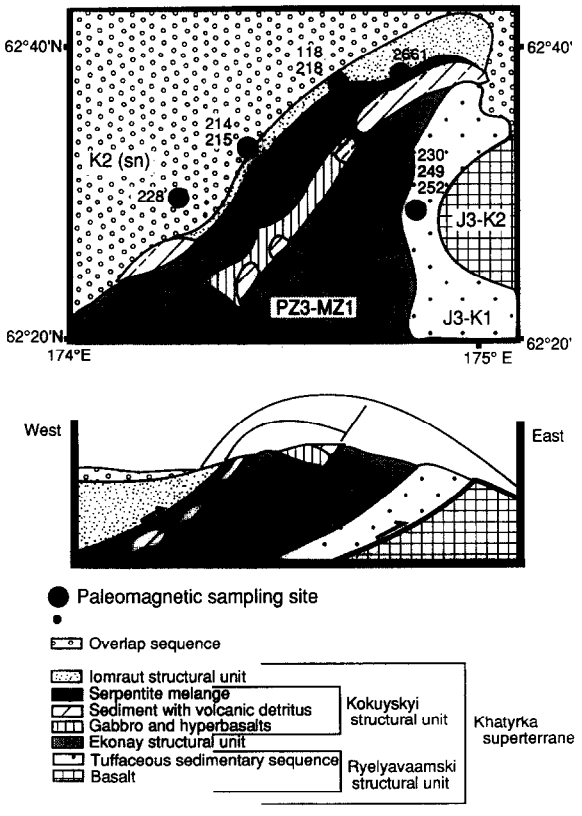


Fig. 3. Geological sketch map the region from which our Khatyrka paleomagnetic samples were collected and a representative cross-section through this area. The Maynitsky paleomagnetic samples were collected from blocks in a sheared and complexly deformed melange zone.

rocks (Sokolov, 1988; Stavsky et al., 1990). The oldest of these are Upper Jurassic to Lower Cretaceous (Neocomian) and of similar lithology to other structural units within the Khatyrka superterrane. Lower Cretaceous (Albian) to Upper Cretaceous strata overlie these older units along a stratigraphic or, rarely, angular unconformity. Maastrichtian to Paleocene coal-bearing molasse rocks are, in turn, discordant with respect to nearby Lower to Upper Cretaceous sediments. The gradual decrease in deformation up section is a characteristic feature of the Velikorechensky zone.

Description of local geology

Paleomagnetic localities 230, 249 (sites 1 and 2), and 252 were collected from well-bedded tuffaceous sandstone members of the Ryelyavaamski structural unit of the Khatyrka superterrane. This unit consists of several fault bounded slices composed of tuffaceous terrigenous sediments, siliceous and volcanic rocks of various ages, including Tithonian, Neocomian and Senonian. These ages are derived from radiolarians found in the cherts. The most widely distributed formations in this structural unit are the tuffaceous sediments, which are characterized by relics of the Upper Jurassic to Valanginian radiolaria. The clastic component of these sediments consists of fragments of basalts, andesites, felsites, granitoids, schists, marbles, and mineral grains of quartz, plagioclase, biotite, epidote and apatite. Ash material constitutes 1–50% of the rock volume. Occasionally the sedimentary matrix consists of a poorly compacted clay–siliceous or clay–carbonate mass enriched with hematite. Igneous flows compose stratigraphic zones, which alternate with jaspers, red and black siliceous shales, pillow basalts and tuffs.

Sampling localities 223 (sites 1 and 2) and 1412 (sites 1, 2, and 3) were taken from thin, tuff–terrigenous members of the Triassic portion of the Kokuyskyi structural unit of the Khatyrka superterrane. This unit is composed of blocks in a serpentinite matrix. Sheared serpentinites make up approximately half of the groundmass of this unit. Of the blocks within the sheared matrix, approximately 3/4 are composed of siliceous rocks of the underlying Ekonay structural unit and terrigenous rocks of the overlying Iomraut structural unit. Rare ultrabasic, gabbro and diabase rocks are also found within the serpentinite matrix. Along the whole belt of the serpentinite melange, there are blocks up to 4 km in size composed of volcanic flows and volcanoclastic sedimentary lithologies of Upper Triassic age.

Tuffs, tuffaceous sandstones, flows of pillow basalts, andesites, and major bodies of intrusive rhyolites are observed in and near this serpentinite zone. The age of this volcanic and sedimentary sequence is Upper Triassic, determined on the basis of bivalves and radiolarians found in the fine-grained tuffaceous sandstones. The presence of the combination basalt, andesite and rhyolite has been interpreted to represent a volcanic se-

ries associated with one or several island-arc terranes.

A series of tectonic outliers of the Kokuyskyi structural unit rocks has been mapped southeast of the serpentinite melange belt. Two slabs overlying intensively deformed upper Palcozoic cherts have been mapped. The lower slab is composed of a complex of foliated gabbro, harzburgites and lherzolites. The upper slab consists mainly of units of rhyolites in association with the Triassic volcanic and sedimentary rocks.

The Iomraut structural unit has been overthrusted onto the serpentinite melange at an angle of 45–60°, dipping to the northwest (Fig. 3). This structural unit is composed of Upper Jurassic to Lower Cretaceous (Neocomian) calcareous, thinly bedded, tuffaceous sandstones, and thinly bedded, tuffaceous turbidites. With shell detritus and oyster shoals, these sedimentary rocks are generally graywacke in composition; arkosic material is not observed to exceed 10–20%.

The Ekonay structural unit is mainly composed of upper Paleozoic to Triassic siliceous rocks. Approximately 90% of this unit is composed of sheared thinly bedded jaspers, cherts, clay-rich sandstones and lens-like bodies of limestones and basalt flows between 5 and 50 m thick (rarely up to 700 m in thickness). The basalts are characterized by a high content of Ti, Fe, P, interpreted to represent oceanic island tholeiites (Ruzhentsev et al., 1982).

Paleomagnetic localities 214 (sites 1 and 2), 215 (sites 1 and 2), and 228 were collected from turbidites and fine-grained sandstones (characterized by the presence of the remains of bivalves) from an overlapping sedimentary sequence. The overlap sequence is composed of Senonian and Paleocene lithic terrigenous and tuffaceous sandstones. These units are composed of alternating coarse and medium clastic tuffs, tuffaceous sandstones, turbidites and tuffstones. Pyroclastic cement is composed of fragments of volcanic glass and small angular pieces of plagioclase, quartz, potassium feldspar, biotite, granites, metamorphic schists and shales, gneisses, argillites, basalts, andesites and rhyolites.

On the whole, the section of the Senonian/Paleocene deposits has been interpreted to represent a marine molasse deposit; cross-bedding, abundance of wood remains, and rapid succession of facies are all characteristic features of this overlapping sedimentary sequence.

From the Yagyelnyi tectonic zone, within the Maynitsky superterrane, a zone which includes gabbros, ultrabasics, dike complexes and basalt flows, located to the northwest of the Khatyrka superterrane, we collected paleomagnetic locality Y1 (site 3). Sites Yi-1, Yi-2, Yi-3, and Yi-4 were collected from gabbro and basalt layers, from blocks within a sheared matrix, of Upper Jurassic to Lower Cretaceous age.

Paleomagnetic techniques and results

We collected oriented hand samples in locations where bedding was clearly visible and easily measured. Paleomagnetic sampling localities are within the Khatyrka and Maynitsky superterranes and a sedimentary sequence which overlaps the Khatyrka superterrane (Fig. 3). In order to average the secular variation of the geomagnetic field, a minimum of five samples were collected at several stratigraphic levels at each locality. All hand sample orientation was done using a magnetic compass and the correction for magnetic variation from geographic north was taken from topographic maps of the sampling area.

After cutting the oriented blocks into samples consisting of 20 mm cubes, all four collections were transported to the magnetically shielded room at the Pittsburgh University Paleomagnetic Laboratory (maximum field in storage region $<$ 100 nT), where paleomagnetic analyses were carried out.

Paleomagnetic measurements of the samples were made using a 3-component cryogenic magnetometer. A Schonsted magnetically shielded furnace was used to complete step-by-step thermal demagnetization within the 200–640°C range. The magnetic field in the cooling region of this furnace varied between 0 and 3 nT.

We divided our samples into three groups on the basis of their magnetic direction and intensity characteristics during stepwise thermal demagnetization. The first group consists of samples which showed a linear demagnetization trend between

thermal demagnetization steps of 400°C and 600°C. Unfortunately, this group accounts for only about 10% of our 140 studied samples. The second and most numerous group of our entire collection, displays a linear demagnetization trend between thermal steps of ~400°C and 530°C (Figs. 4 and 5). In order to calculate a characteristic magnetization for these samples we have used principal component analysis to calculate a best-fitting line to demagnetization data within the temperature interval over which a single, high-unblocking temperature, magnetic component was removed. Thermal demagnetization steps from 550°C and higher cause these samples to show a dramatic increase in magnetization and display directional instability between heating steps. All four study localities have some samples which behave in this manner. The third group displayed a more complex behavior during demagnetization (Fig. 6).

Both of the first two groups reveal two magnetic components, a characteristic higher-temperature unblocking magnetic component and a lower-temperature unblocking magnetic component. The direction of the lower-temperature unblocking magnetic component is well constrained by principal component analysis of the thermal demagnetization data. The initial demagnetization of sample 215/5, shown in Figure 5, during thermal demagnetization steps between NRM and 300°C shows the removal of a low-temperature unblocking magnetic component, which has a steep downward directed inclination and southward declination. Analysis of thermal demagnetization data for 30 samples stepwise heated between 20°C and 300 or 400°C, gives an estimate

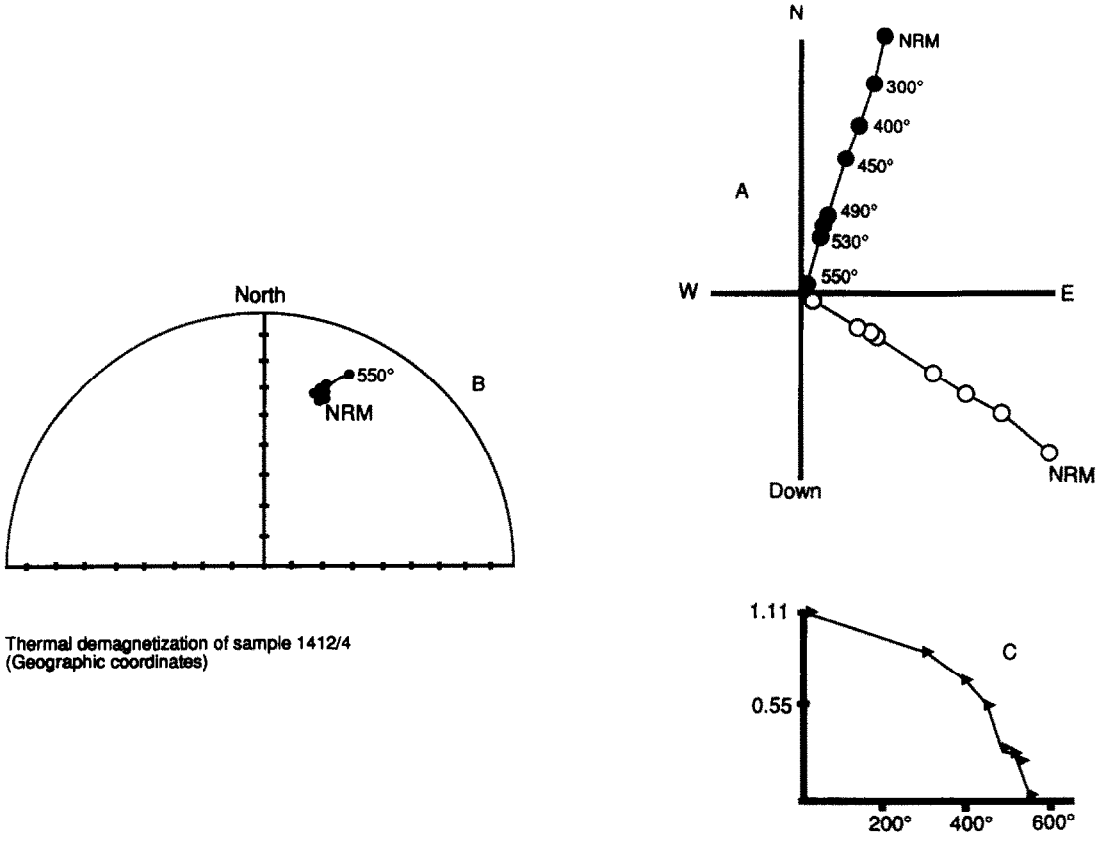


Fig. 4. Representative thermal demagnetization diagrams for sample 1412/4. (A) Orthogonal vector plot, • = projection onto horizontal plane, ○ = projection of paleomagnetic demagnetization data onto vertical plane. (B) Equal area, lower hemisphere, stereographic projection of directions of magnetization for this sample, solid circles represent projections onto the lower sphere. (C) Intensity vs. thermal demagnetization step diagram for this sample, triangles represent data points.

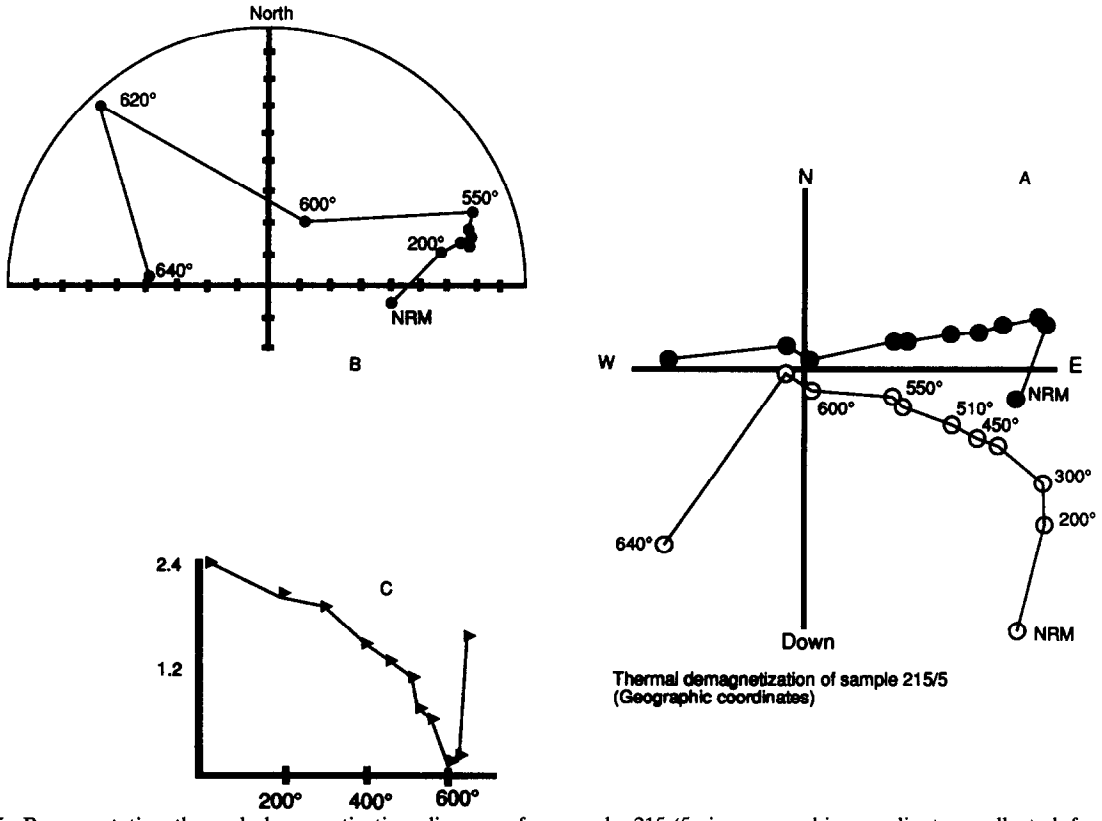


Fig. 5. Representative thermal demagnetization diagrams for sample 215/5, in geographic coordinates, collected from the sedimentary overlap sequence of the Khatyrka superterrane. (A) Orthogonal vector plot, • = projection onto horizontal plane, ○ = projection of paleomagnetic demagnetization data onto vertical plane. (B) Equal area, lower hemisphere, stereographic projection of directions of magnetization for this sample, solid circles represent projections onto the lower sphere. (C) Intensity vs. thermal demagnetization step diagram for this sample, triangles represent data points. Note normal polarity, low unblocking temperature magnetic component which is removed during thermal demagnetization steps between 300–200°C and NRM.

of the low-temperature unblocking magnetic component of $D = 1°$, $I = 75°$, $\kappa = 31$ and α-95 = 4.6°, in geographic coordinates similar to the present-day geomagnetic field at the sampling site (present-day field: $D = 3°$, $I = 72°$). We interpret this lower-temperature unblocking magnetic component to represent an overprint of the present-day magnetic field on these samples.

In terms of defining a higher-temperature unblocking magnetic component, after final analysis approximately 1/3 of all of our samples had to be rejected because they displayed no coherent characteristic remanent magnetization during thermal demagnetization. Samples from localities 218 and 266 were entirely rejected as these displayed chaotic directional and magnetic intensity behavior during stepwise thermal demagnetization.

Paleomagnetic site-mean directions were more closely grouped in stratigraphic coordinates. The Fisher estimate of κ for the sites collected with in the Yagyelnyi structural unit of the Mayniski superterrane increased from 8 to 26 ($N = 5$, ratio 3.3) after correcting for the tilt of the sampled units. Similarly, after correcting for tilt, the dispersion of the sites collected from the Kokuyskyi structural unit of the Khatyrka superterrane decreased, with the Fisher estimate of κ increasing from 6 to 26 ($N = 5$, ratio 4.3). The Ryelyavaamski structural unit sites also were better grouped in stratigraphic coordinates with the Fisher estimate of κ increasing from 8 to 110 ($N = 5$, ratio 13.8). In addition, the collection made from the sedimentary sequence overlapping the Khatyrka superterrane also were significantly better

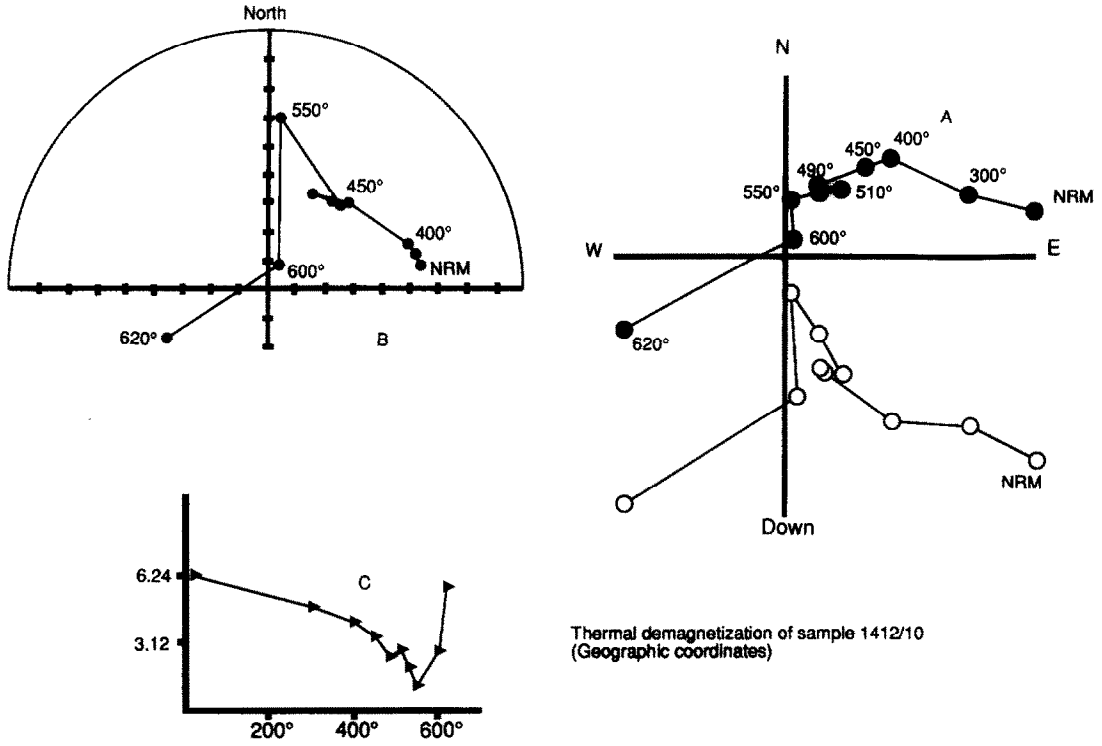


Fig. 6. Representative thermal demagnetization diagram in geographic coordinates, collected from the Kokuyskyi structural unit of the Khatyrka superterrane (sample 1412/10; from paleomagnetic site 1410/3). (A) Orthogonal vector plot, ● = projection onto horizontal plane, ○ = projection of paleomagnetic demagnetization data onto vertical plane. (B) Equal area, lower hemisphere, stereographic projection of directions of magnetization for this sample, solid circles represent projections onto the lower sphere. (C) Intensity vs. thermal demagnetization step diagram for this sample, triangles represent data points.

grouped when corrected for the tilt of the sampled units as shown by the increase in κ from 7 to 123 ($N = 5$, ratio 15.4).

All of our collections pass fold tests at 95% confidence using the critical κ ratio value given by McElhinny (1964). The overlap sequence, Ryelyavaamski, and Yagyelnyi results pass fold tests at 99%. Partial unfolding tests were completed for each structural unit (Fig. 7). Two of the structural units studied, Ryelyavaamski and

TABLE 1

Characteristic site-mean directions for rocks of the Khatyrka terrane sedimentary overlap sequence

Number	Sites	Bedding	Azimuth	Characteristic magnetic direction				
				Geographic		κ/α-95	Stratigraphic	
				D	I		D	I
214/1	$n = 4$	288°	33°	97°	65°	350/4°	317°	80°
214/2	$n = 5$	288°	13°	68°	75°	155/5°	10°	80°
215/1	$n = 6$	260°	60°	81°	33°	158/5°	244°	87°
215/2	$n = 3$	240°	55°	57°	23°	326/5°	47°	78°
228	$n = 13$	25°	20°	205°	74°	98/4°	25°	86°
$N = 5$				78.7°	59.9°		5.5°	84.4°
					6.2/33.3°		122.8/6.9°	

n = number of samples; N = number of sites; κ and α-95 are Fisherian precision parameters. Sites 214/1, 214/2, 215/1 and 215/2 were collected from different stratigraphic levels.

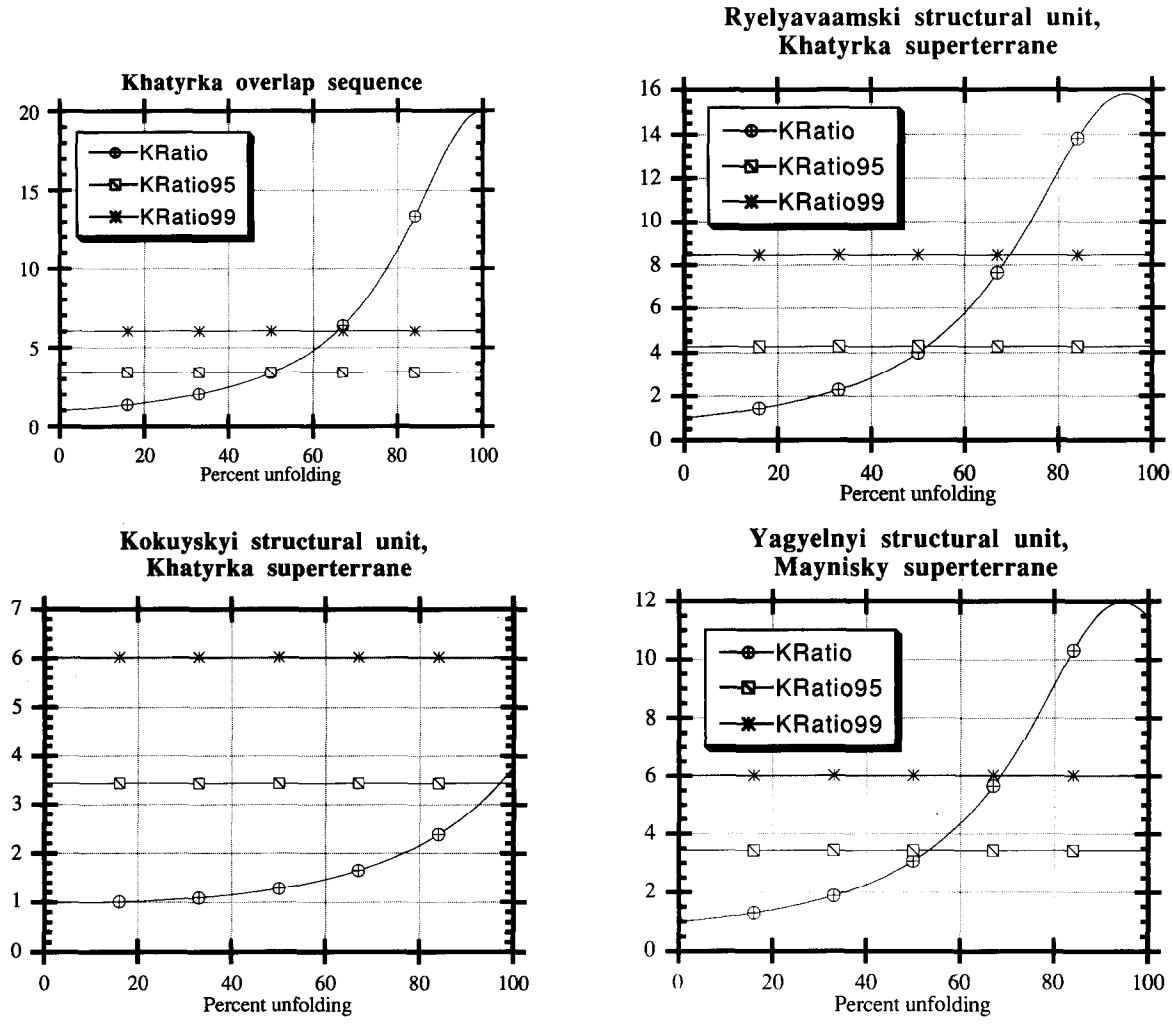


Fig. 7. Partial unfolding of all sampling regions. Shown as a function of percent unfolding are the Fisher estimate of κ and the values for significance of the fold test at 95 and 99% confidence (from McElhinny, 1964). (A) Khatyrka overlap sequence, passes fold test at 99% confidence; (B) Ryelyavaamski structural unit, Khatyrka superterrane, passes fold test at 99% confidence; (C) Kokuyskyi structural unit, Khatyrka superterrane, passes fold test at 95% confidence; and (D) Yagyelnyi structural unit, Maynitsky superterrane, passes fold test at 99% confidence. Folding of sampled units was complex and we could not apply the McFadden and Jones (1981) revised fold test to our data.

Yagyelnyi, show maximum κ values at 94% unfolding. However, the small change in κ and direction do not differ significantly when compared with the corresponding data at 100% unfolding.

The paleomagnetic samples collected from the sedimentary overlap sequence are observed to have only a normal polarity (Fig. 8; Table 1). However, the positive fold test suggests that the characteristic magnetization is primary. We interpret the consistent normal polarity in these samples as due to the geomagnetic field during deposition, which was of normal polarity during more than 75% of the Upper Cretaceous period. Two other studied intervals (the Iomraut structural unit: Carnian–Norian; the Ryelyavaamski structural unit: Kimmeridgian–Valanginian) are within mixed polarity zones.

The magnetization of the Relyevaamian structural unit sedimentary rocks are mostly of downward directed magnetic polarity (Table 2). However a single sample has a characteristic magnetization of $D = 223°$, $I = -47°$ and was collected at the lowest stratigraphic level of the site 249/2

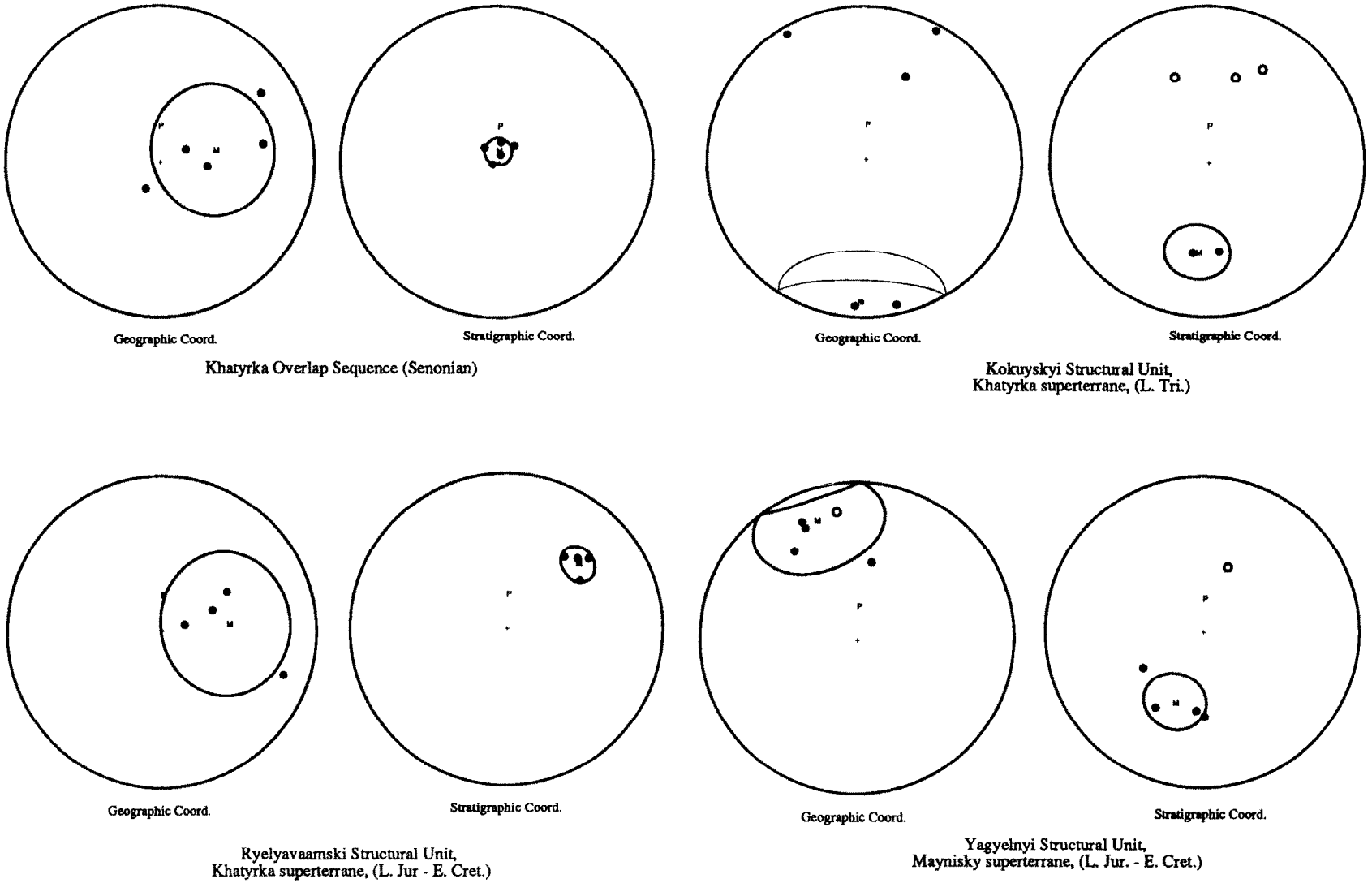


Fig. 8. Distribution of characteristic site-mean magnetic directions plotted on an equal area sterographic projections. (A) Khatyrka overlap sequence; (B) Kokuyskyi structural unit; (C) Ryelyavaamski structural unit; and (D) Yagyelnyi structural unit. The α-95 is shown about the average direction, which is labelled with *M*. *P* is the location of present-day magnetic north ($I = 3°$, $D = 72°$) in the area sampled. Solid circles and squares represent lower hemisphere projections, open symbols represent projection onto the upper hemisphere.

TABLE 2

Characteristic site-mean directions for rocks of the Ryelyavaamski structural unit

Number	Sites	Bedding	Azimuth	Characteristic magnetic direction				
				Geographic		κ/α-95	Stratigraphic	
				D	I		D	I
230	$n = 6$	25°	20°	59°	50°	31/11°	50°	32°
249/1	$n = 6$	50°	35°	74°	78°	103/6°	57°	44°
249/2	$n = 4$	5°	30°	69°	62°	61/9°	39°	41°
252	$n = 10$	340°	92°	110°	18°	69/6°	46°	37°
$N = 4$				83.7°	54.5°		47.9°	38.7°
					7.6/35.6°		109.9/8.8°	

n = number of samples; N = number of sites; κ and α-95 are Fisherian precision parameters. Sites 249/1 and 249/2 were collected from different stratigraphic units. Highest κ is at 94% unfolding $D = 49.3°$, $I = 40.5°$, $\kappa = 120.7$, and α-95 = 8.4°. These data do not differ significantly from the values at 100% unfolding.

outcrop. When reflected through the origin it is well grouped with the remaining normal polarity samples from this locality.

The samples collected from the Kokuyskyi structural unit of the Khatyrka superterrane have two magnetic polarities: a downward directed magnetic polarity within the sediments and a upward directed magnetic polarity in the basalt flows (Fig. 8; Table 3). Figure 8 shows a zonal dispersion of the magnetizations in the Kokuyskyi structural unit sites where inclinations are similar between samples but the declinations differ by more than 50°. These samples were collected from blocks in a melange matrix and we interpret this spread in declination to represent post-accretional relative rotation of the sampled blocks.

We see a similar dispersion of characteristic paleomagnetic declinations, but not of the characteristic inclinations in our study of the island arc fragments within the Maynitsky superterrane (Fig. 8; Table 4). This zonal disposition is probably caused by blocks located in the serpentinite melange turning about individual axes of rotation.

TABLE 3

Characteristic site-mean directions for rocks of the Kokuyskyi structural unit

Number	Sites	Bedding	Azimuth	Characteristic magnetic direction				
				Geographic		κ/α-95	Stratigraphic	
				D	I		D	I
223/1bas	$n = 5$	305°	55°	168°	6°	221/6°	189°	41°
223/2sed	$n = 7$	300°	57°	328°	7°	29/12°	339°	−42°
1412/1bas	$n = 4$	33°	42°	184°	7°	318/6°	173°	42°
1412/2sed	$n = 5$	55°	80°	26°	42°	30/15°	30°	−32°
1412/3sed	$n = 5$	55°	55°	29°	5°	64/10°	18°	−43°
$N = 5$				181.5°	−8.6°		186.5°	41.5°
					6.4/32.8°		25.5/15.4°	
				181.3°	−6.9° *		186.5°	41.5° *
					(−8.2, −3.8,61°		(−377.8, −8.4,78°,	
					18.5°, 29.2°)		2.2°, 14.9°)	

n = number of samples; N = number of sites. Lithology sampled: bas = basalt, sed = sedimentary. κ and α-95 are Fisherian precision parameters.

* Mean directions were calculated using Bingham statistics (error parameters: κ-1, κ-2, error ellipse azimuth, α-95-1, α-95-2). Sites 1412/2 and 1412/3 were collected from distinct stratigraphic units.

TABLE 4

Characteristic site-mean directions for island arc rocks of Yagyelnyi structural unit, Maynitsky terrane

Number	Sites	Bedding	Azimuth	Characteristic magnetic direction					
				Geographic		κ/α-95	Stratigraphic		κ/α-95
				D	I		D	I	
Yl-3/bas	$n=14$	187°	83°	10°	49°	14/10°	184°	48°	31/7°
Yi-l/bas	$n=4$	317°	50°	351°	−19°	66/9°	21°	−54°	
Yi-2/gab	$n=6$	165°	115°	335°	19°	17/14°	178°	45°	46/9°
Yi-3/gab	$n=6$	175°	95°	325°	32°	48/8°	211°	43°	
Yi-4/gab	$n=6$	185°	90°	335°	21°	31/9°	238°	54°	78/6°
$N=5$				341.7°	21.4°	8.0/28.8°	201.2°	50.7°	25.9/15.3°
				340.4°	22.4° *		200.9°	50.6° *	
					(−13.1, −4.3,1,		(−81.8, −9.3,76°,		
					13.9°, 25.8°)		4.6°, 14.2°)		

n = number of samples; N = number of sites. Lithology sampled: bas = basalt, sed = sedimentary, gab = gabbro. κ and α-95 are Fisherian precision parameters.

* Mean directions were calculated using Bingham statistics (error parameters are κ-l, κ-2, error ellipse azimuth, α-95-1 and α-95-2). Highest κ is at 94% unfolding, $D=202.9°$, $I=54.6°$, $\kappa=27.8$, α-95 = 14.8°. These data are not significantly different from those at 100% unfolding.

TABLE 5

Summary of characteristic magnetic poles

Locality	Age	Sampling location	Mean pole [1]			Mean pole [2]			Mean Pole [3]							Paleo latitude
			λ	ϕ	dm,dp	λ	ϕ	α-95	λ	ϕ	κ-1	κ-2	Az.	α-1	α-2	
Khatyrka overlap seq.	Senonian	62.5°N, 174.4°E	73.5°	178.1°	13.6°,13.4°	73.3°	177.3°	13.6°	73.3°	177.3°	−35.8	−14.7	67°	7.0°	11.0°	79°
Khatyrka superterrane Ryelyavaamski st. ut.	L. Jur.–E. Cret.	62.4°N, 174.8°E	38.2°	293.6°	10.5°,6.2°	38.2°	293.5°	8.9°	38.2°	293.6°	−159.2	−46.3	109°	3.2°	6.0°	22°
Maynitsky terrane Yagyelnyi st. ut.	L. Jur.–E. Cret.	63.2°N, 175.3°E	5.5°	156.4°	20.6°,13.9°	6.4°	156.8°	20.1°	6.2°	157.3°	−106.5	−5.3	71°	4.1°	19.6°	32°
Khatyrkasuperterrane Kokuyskyi st. ut.	L. Triassic	62.5°N, 174.5°E	−2.8°	169.4°	18.8°,11.5°	−3.4°	168.8°	18.2°	−3.4°	168.7°	−682.0	−6.1	94°	1.6°	17.9°	24°

[1] Calculated by converting average directions to pole position.

[2] Calculated by converting directions to pole positions and calculating Fisher average.

[3] Calculated by converting directions to pole positions and calculating Bingham average.

Discussion of results

Positive paleomagnetic reliability tests (Coe et al., 1985; Van der Voo, 1988; Harbert, 1990) must be present in any paleomagnetic study before it may be tectonically interpreted. These tests include the observation that the characteristic remanent magnetizations are dissimilar to either the present-day magnetic field or the most likely overprint magnetic field, positive fold tests, and the presence of more than one polarity of characteristic remanent magnetization. In our study, the most likely secondary magnetic component is a Cretaceous steep-inclination direction, corresponding in time to the activity of the nearby Okhotsk–Chukotsk plutonic belt, or a steep present-day field direction. Both of these differ significantly from our observed high-temperature characteristic magnetic components (Fig. 8). In addition, we observed two polarities of characteristic remanent magnetizations and our sites pass, at 95/99%, fold tests. We therefore interpret the high-temperature unblocking magnetic component observed as the characteristic remanent magnetization acquired during or shortly after deposition of the sampled units. We interpret these characteristic remanent magnetizations by converting them to virtual geomagnetic poles (VGP) and calculating paleolatitudes of deposition for each of the sampled units. We calculate two VGP's from the Khatyrka superterrane, a VGP for the Maynitsky superterrane, and a VGP from the Senonian sedimentary overlap sequence (Table 5). Each of the paleolatitudes from the Khatyrka and Maynitsky superterranes are anomalously low (Fig. 9; Table 5). The paleomagnetic pole from the Senonian sedimentary sequence suggests deposition of this unit at high paleolatitude.

Within the Khatyrka superterrane the observed paleomagnetic paleolatitude is 24°N or S for the island arc complex of the Upper Triassic Kokuyskyi structural unit. The observed paleomagnetic paleolatitude from the Jurassic–Creta-

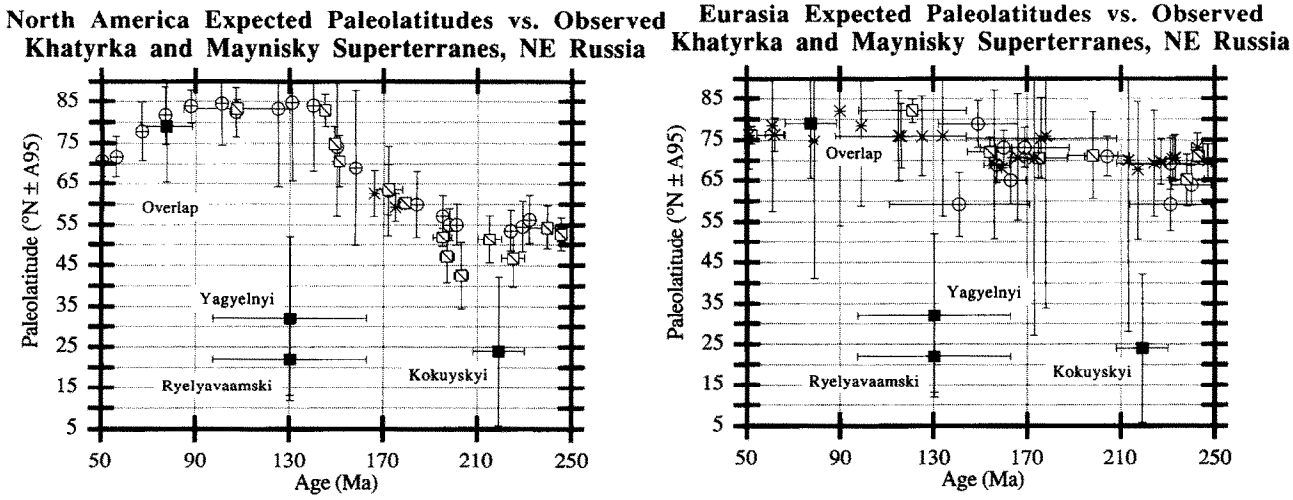


Fig. 9. Comparision of the results of our paleomagnetic reconnaissance studies with expected paleolatitudes taken from North America and Europe. The paleolatitudes of our studies are shown as solid squares. (A) North America: expected paleolatitude for the Koryak region from the North America apparent polar wander path of Irving and Irving (1982) (○), May and Butler (1986) (□), Van Fossen and Kent (1990) (×) and Witte and Kent (1989) (we used their Newark B magnetic component: ×). (B) Europe: expected paleolatitude for the Koryak region from the Europe high-reliability studies tabulated in Frei and Cox (1987) (○), pole of Halvorsen (1989) (○), pole of Johnson et al., (1984), the European apparent polar wander path of Westphal et al. (1986) (□), Irving and Irving (1982) (×), and Khramov (1987) (×). With respect to expected paleolatitudes, the overlap sequence agrees well with that expected, which suggests these units were in-place by Senonian. Older units appear to have formed at significantly shallower paleolatitudes. We cannot constrain the hemisphere of deposition, thus the Yagyelnyi, Ryelyavaamski, and Kokuyskyi could represent northern or southern hemisphere paleolatitudes; either choice places them significantly south of their expected positions on either the North America or Eurasia plates.

ceous, Ryelyavaamski structural unit of this superterrane is 22°N or S. Both of these paleolatitudes are consistent with each other but differ significantly from that expected for either the Eurasia or North America plates. These results strongly suggest that the Khatyrka superterrane is both far travelled and allochthonous with respect to this region of the northeastern Russia. We cannot, on the basis of these data alone, specify whether these paleolatitudes represent northern or southern hemisphere latitudes of deposition. Either interpretation, when compared with expected paleolatitudes for the North America or Eurasia plates, show substantial northward motion of this superterrane.

The ophiolite rocks studied in the Maynitsky superterrane also differ significantly in paleolatitude when compared with that expected for either the Eurasia or North America plates. The paleolatitude calculated from the Yagyelnyi structural unit of the Maynitsky superterrane yields a paleolatitude of 32°N or S, significantly shallower than expected (Fig. 9; Table 5).

The paleomagnetic data for the Senonian overlap sediments of the Khatyrka superterrane suggest that these sediments were deposited at high paleolatitude, along the southeastern edge on the Eurasia plate (Table 5). There is no significant difference between the expected and observed paleolatitudes. However, there is a 20° clockwise rotation of the observed magnetic declination away from that expected declination for a site deposited on the Eurasia plate during the Senonian. Because very close to the pole, the observed rotation is not significant with respect to the uncertainties of this study and the reference pole used.

Conclusion

We have completed a reconnaissance paleomagnetic study of units within the Khatyrka superterrane, Maynitsky superterrane and a Senonian overlap sequence. Our results show that the Ryelyavaamski and Kokuyskyi structural units of the Khatyrka represent far-travelled superterranes. In addition our paleomagnetic results from the Yagyelnyi structural unit of the Maynitsky superterrane also show that this superterrane is allochthonous. The paleomagnetic results from the Senonian sedimentary overlap sequence show that these units were deposited in-place with respect to their present position in northeastern Russia.

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